Spatial Data Quality Dashboard Design

1. Purpose

The purpose of the dashboard is to monitor and assess the quality of spatial data in a clear and efficient way. It enables users to quickly identify errors, detect outliers, and evaluate whether the data is suitable for its intended use. By providing a structured overview, the dashboard supports better decision-making and helps improve data reliability.

2. Target Users

The dashboard is designed for multiple types of users. GIS analysts and data producers use it to identify and correct errors in the dataset. Quality managers monitor overall data quality performance, while decision-makers use the dashboard to evaluate whether the data is reliable and fit for purpose.

3. Metrics

The dashboard includes key spatial data quality indicators. These include positional accuracy measures such as RMSE and accuracy at the 95% confidence level, as well as the number and percentage of outliers. In addition, other important quality elements are considered, including attribute accuracy, logical consistency, completeness, and temporal accuracy. These metrics provide a comprehensive view of data quality. These metrics are aligned with the ISO 19157 data quality framework, including positional accuracy, thematic accuracy, completeness, and temporal accuracy.

4. Dashboard Content

The dashboard is structured to present both summary information and detailed insights. A KPI section at the top provides a quick overview of key indicators such as RMSE, accuracy, and outliers. Below this, the distribution of errors is visualized to show how values are spread and to highlight extreme deviations. A spatial component allows users to understand where errors occur geographically. Additionally, a table is included to list the largest errors and clearly identify outliers. Table 1 illustrates the proposed layout of the dashboard, showing the arrangement of key performance indicators (KPIs), error distribution chart, spatial map, and outlier table.

Table 1: Proposed Spatial Data Quality Dashboard Mockup

Spatial Data Quality Dashboard (Balzan, Malta)		
RMSE: 0.203 m	Accuracy: 0.35 m	Outliers: 2.5%
Filters: [Dataset ▼] [Date ▼]		
Error Distribution (Histogram)	Spatial Error Map	
Point Error Status MAP024 8.0 m Outlier MAP031 0.35 m Normal		

5. Design Choices

The dashboard is designed to be simple, clear, and easy to interpret. A clean layout is used to avoid unnecessary complexity. Color coding is applied to highlight data quality levels, with green indicating acceptable values and red indicating critical issues. Key indicators are placed at the top to allow immediate understanding, while more detailed visualizations are positioned below to support deeper analysis.

6. Justification of Design

The design supports both quick assessment and detailed analysis. Users can immediately evaluate overall data quality through the KPI panel, while additional visualizations help them understand patterns and distributions of errors. The combination of statistical indicators and visual elements ensures that both technical and non-technical users can effectively interpret the data.

7. Outlier Analysis

The analysis shows that most error values are concentrated around zero, indicating generally good positional accuracy. However, a small number of outliers with significantly larger errors were identified. These extreme values have a noticeable impact on overall RMSE and accuracy. This highlights the importance of detecting and analyzing outliers when assessing spatial data quality, as they can strongly influence the results.

8. Conclusion

The proposed dashboard provides an effective tool for monitoring and improving spatial data quality. By combining key indicators, visual representations, and clear organization, it allows users to quickly assess data reliability and identify potential issues. The design ensures that the dashboard is both practical and informative, supporting efficient quality control and better decision-making.